

Generation Interconnection System Impact Study Report

For Queue Project Q21-1728

Inverness 138 kV

May 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1728. The Interconnection Customer proposes a 115.8 MW solar photovoltaic facility interconnecting at the Inverness 138 kV substation in the Pennsylvania area. The request was received on April 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 393 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,930,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W Gallatin 345/138kV Autotransformer
Contingency: West Gallatin / Felida 345.0 kV line
Loading: 103.7% (Rating: 480 MVA)
DFAX: 8.1%
Pre-project loading: 101.7%
2. Facility: Felida / Satsop 138.0 kV line
Contingency: Stsop Substation / Meredosia 138.0 kV
Loading: 101.9% (Rating: 241 MVA)
DFAX: 28.2%
Pre-project loading: 88.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Gallatin 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$13,253,000	14% (\$1,845,000)
Felida / Satsop 138.0 kV line	Reconductor line	\$549,000	16% (\$85,000)
	Total allocated to project		\$1,930,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.